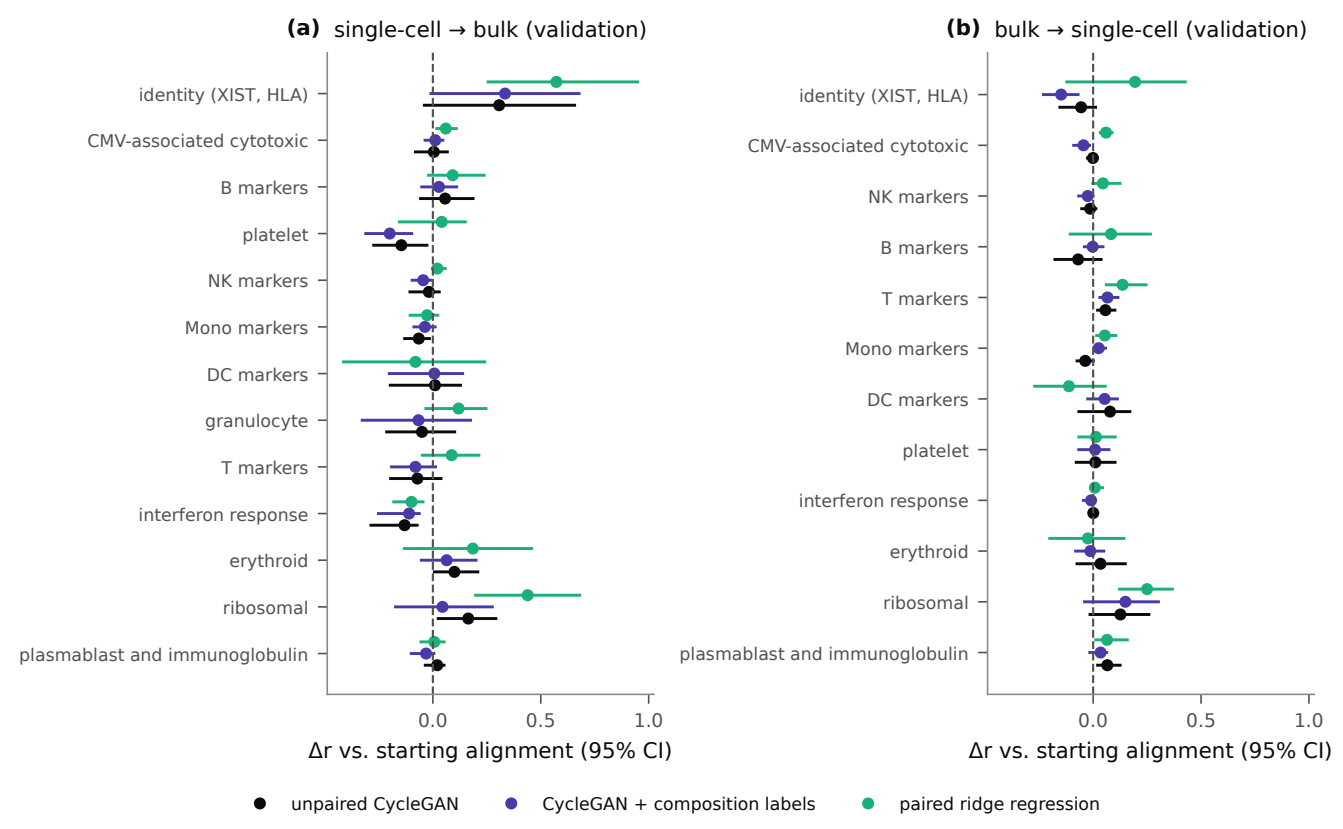
(a) single-cell → bulk (validation)
identity (XIST, HLA)
CMV-associated cytotoxic
B markers
platelet
NK markers
Mono markers
DC markers
granulocyte
T markers
interferon response
erythroid
ribosomal
plasmablast and immunoglobulin
0.0
0.5
1.0
Δr vs. starting alignment (95% CI)
(b) bulk → single-cell (validation)
identity (XIST, HLA)
CMV-associated cytotoxic
NK markers
B markers
T markers
Mono markers
DC markers
platelet
interferon response
erythroid
ribosomal
plasmablast and immunoglobulin
0.0
0.5
1.0
Δr vs. starting alignment (95% CI)
unpaired CycleGAN
CycleGAN + composition labels
paired ridge regression